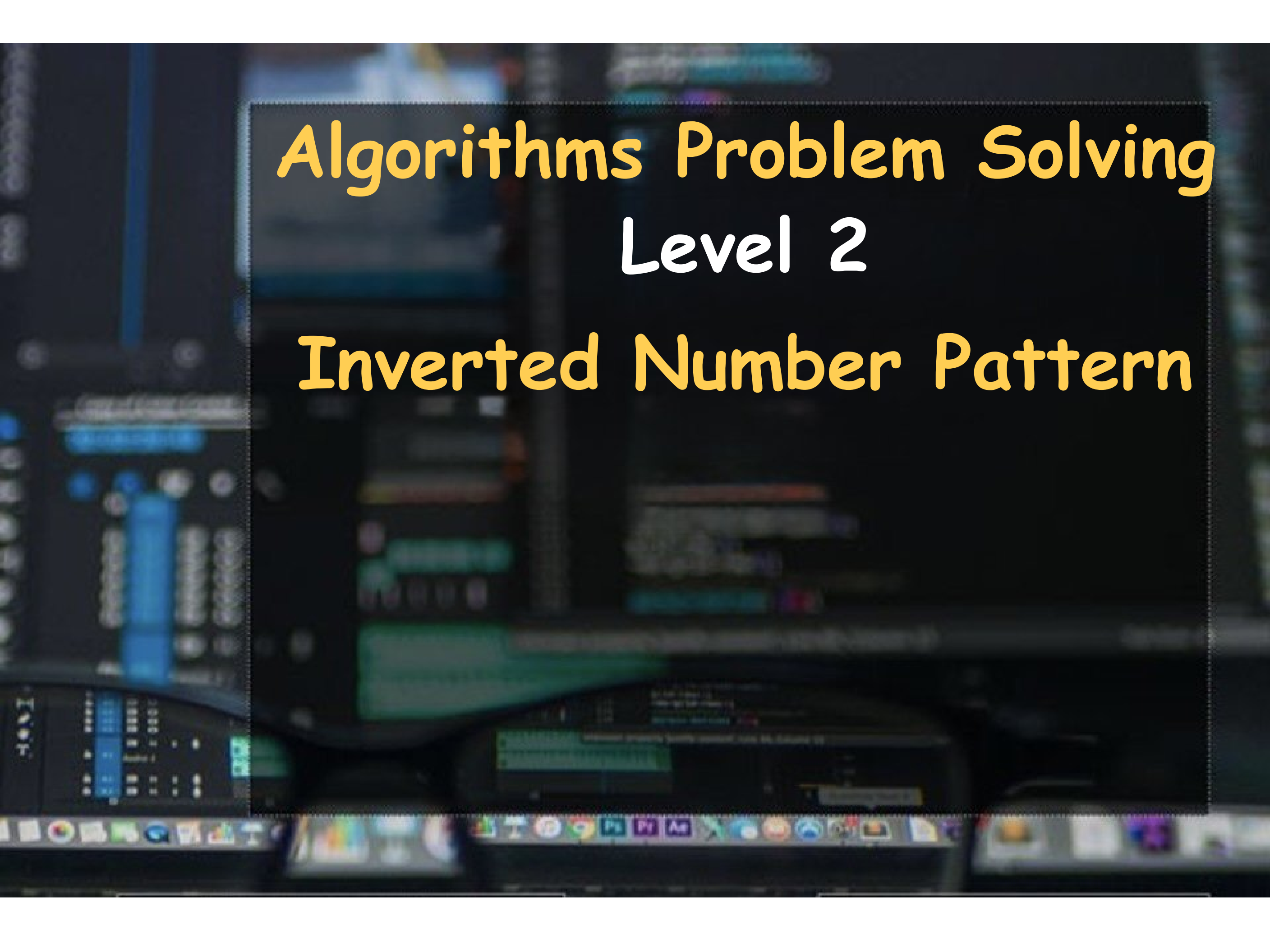
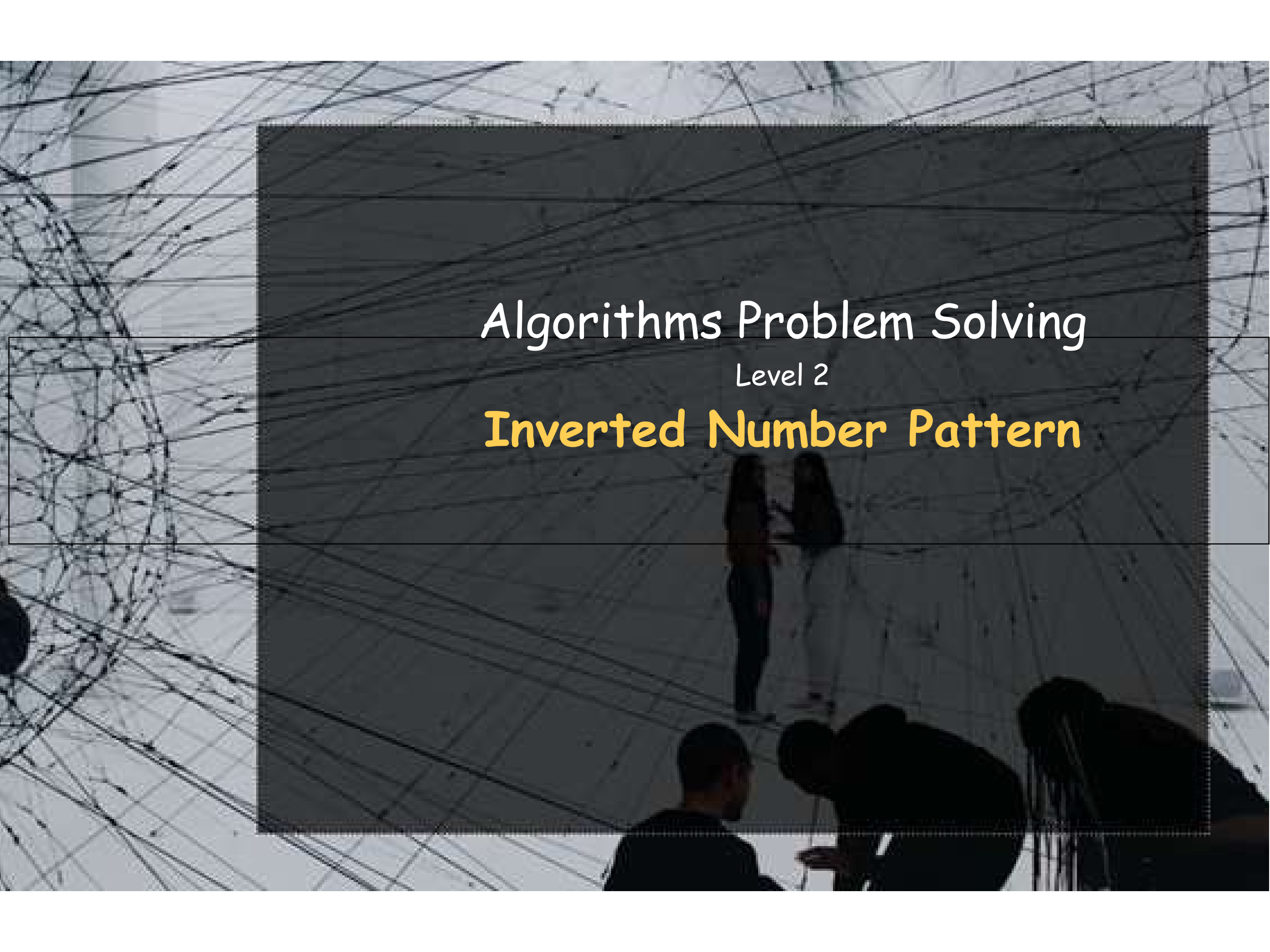




Algorithms Problem Solving Level 2

Inverted Number Pattern



Algorithms Problem Solving

Level 2

Inverted Number Pattern

Definition

- ◆ An inverted number pattern is a pattern where the number of printed digits decreases row by row.
- ◆ The program starts from the entered number and decreases the number by `1` after each row.
- ◆ For example, input `3` produces `333`, `22`, `1`.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``PrintInvertedPattern()`` uses two nested ``for`` loops.
- ◆ The outer loop controls the current number and the number of rows.
- ◆ It starts from ``Number`` and continues down to ``1``.
- ◆ The inner loop prints the current value of ``i`` several times.
- ◆ The inner loop runs ``i`` times.
- ◆ After the inner loop finishes, ``endl`` moves to the next line.
- ◆ When ``i`` decreases, both the number being printed and the number of repetitions decrease.

Steps in main()

1. Read a positive number.
2. Start the outer loop from `Number` down to `1`.
3. Start the inner loop from `1`.
4. Print `i` while `j <= i`.
5. Move to the next line.
6. Decrease `i` and repeat.

💡 Example

◆ Input:

``5``

◆ First row: ``i = 5`` → print ``5`` five times.

◆ Second row: ``i = 4`` → print ``4`` four times.

◆ Third row: ``i = 3`` → print ``3`` three times.

◆ Fourth row: ``i = 2`` → print ``2`` two times.

◆ Fifth row: ``i = 1`` → print ``1`` one time.

➡ Result

55555

4444

333

22

1

Note

- ◆ The outer loop controls the rows.
- ◆ The inner loop controls how many times the number is printed in each row.
- ◆ For every row, the value of `i` is used for both the number printed and the number of repetitions.

⊘ Warning

- ◆ The validation allows `0` because the condition is $N < 0$.
- ◆ If the user enters `0`, the outer loop does not run and nothing is printed.



Comparison

▨ Outer Loop

- ✓ Controls which number is printed.
- ✓ Counts down from `Number` to `1`.
- ✓ Example: $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$.

▤ Inner Loop

- ✓ Controls how many times the current number is printed.
- ✓ Runs from `1` to `i`.
- ✓ Example when $i = 4$ → prints `4` four times.

✦ Summary

- ✓ Use an outer loop to count down from `Number` to `1`.
- ✓ Use an inner loop to repeat the current number.
- ✓ The current value of `i` determines both the digit and the repetition count.
- ✓ Each row contains one fewer digit than the previous row.
- ✓ Input `5` produces `55555`, `4444`, `333`, `22`, `1`.

Thank you 😊